Independent same-u UInt64 composition review

The three named declarations passed fresh private Lean checks and source-fidelity review at their stated scope. All 82 producer and dependency bindings matched before and after replay. The main module, copied field fixtures, producer controls, and separate axiom readback each exited zero. Four additional fixed reviewer controls passed after an explicitly retained instance correction. The producer packet was not modified.

Source fidelity: crossWords preserves the retained add22_weighted square/cube/fourth/fifth association and crossed q,r,s,t formulas at projective_law_gate.rs903–925. samePointWords preserves k=s1c−s2c, x=(t2c−t1c)k, y=((s1x+t1(z2²k²))k³)z2³, z=k(z1z2), w=w1. Its field expression bridge is definitional after each arithmetic composition. The source extraction uses no pow call. The subsequent six-field dbl payload matches the retained Rust477–510, including raw literal5 at p=5.

The prime range is $5 \le p < 2^{31}$ with Field.n=1 and tables=None. All twelve coordinate words and four curve words are canonical. For canonical operands, products are below 2^{62} and sums below 2^{32}, hence below 2^{64}; each modular result is canonical before the next operation. Negation/subtraction use the existing source-faithful remainder bridges. The imported doubling bridge separately handles raw5 multiplication, which is not a canonical leaf at p=5. This is a pure value correspondence; mutable counters, their overflow, and table-backed arithmetic are excluded.

Guard composition: support equality is tested before inverse, equal, and zero-slope checks. Canonical equality and negation reflection transport these actual word conditions to the reviewed field theorem. The two original normalized validTwo hypotheses, four nonzero weights and explicit IsCoprime f f.derivative produce nonzero k and extracted ordinate; the theorem does not assume output validity. The doubled payload is canonical, has nonzero z/w, and satisfies normalized validity for the monic quintic $X^5+a3X^3+a2X^2+a1X+a0$. Earlier/native routes return identity or double22; the restricted word dispatcher instead returns None as an explicitly documented scope marker. No theorem certifies those earlier payloads or the whole add22 dispatcher.

Nonvacuity: the full nonunit F5 fixture applies the original-input theorem with a=(0,4,0,1,2,3), b=(0,4,4,0,3,2), coefficients(1,4,0,0), an explicit coprimality certificate, and all literal guards. It verifies k=4, extracted point(1,2,4,3), and doubled words(3,1,2,4,4,4). Producer controls also cover inverse/equal restricted None and a noncanonical coordinate.

Reviewer boundary controls use fixed singular words a=(0,0,1,0,1,1), b=(0,0,2,0,1,1): arithmetic gives k=4, extracted y=0 and raw doubled w=0. A separate checked lemma rejects global coprimality for X^5 over F5. Additional controls check support-mismatch None and p=2 exclusion. These are boundary arithmetic/certificate checks, not a claim that the original-input theorem applies on the singular fixture. All four final assertions pass.

Failures retained: a duplicate wrapper tag was rejected before Lean launched; the first reviewer-control compile lacked Fact(Nat.Prime 5). Its source and compiler output remain alongside the corrected successful check. Producer parse/control failures remain in the immutable producer packet. Axiom readback for all three declarations reports only propext, Classical.choice and Quot.sound. Imported compiled dependencies retain the Lean toolchain trust boundary; no separate kernel implementation was run.

Review disclosure: the reviewer authored the imported SameUComposition, LocalSmooth and GlobalSmooth field components and previously reviewed other imported word bridges. This is an independent source review and fresh replay of the new UInt64 composition, not a blind rederivation of those dependencies. Retained Rust source semantics are bound; current Cargo linkage, divisor class-sum semantics, operational counts and performance are outside the claim. Cost ratio is null. Root retains archival and official promotion authority.